

Patient App — Complete Plan

Companion mobile app for DentOS clinics · Flutter (iOS + Android) · Supabase backend

1 · What we are building

A single patient-facing mobile app, published once on the Play Store / App Store, that serves every clinic running DentOS. A patient links the app to **their** clinic by scanning the clinic's QR code (or typing the clinic ID), then proves their identity by scanning their **personal patient QR** or by entering their **CNIC + full name**. After that they can view and request appointments, receive reminders, and see their visit history — with zero SMS/OTP cost to the clinic.

Why it pays for itself

- **Reminders cut no-shows 30–40%** — push notifications are free (FCM); no-shows are the biggest avoidable revenue leak.
- **Self-booking fills empty slots** without receptionist phone time.
- **Recall alerts bring dormant patients back** for 6-month cleanings — recurring revenue on autopilot.
- **Zero per-login cost** — identity is QR / CNIC based; no SMS provider needed.

2 · Linking & login flow (final design)

Step A — Link to the clinic (mandatory gate)

On first launch the app shows **two options side by side** — the patient chooses either one:

- **Scan clinic QR** — printed at reception, on invoices and appointment cards.
- **Enter clinic ID** — a human-readable code typed manually.

Clinic ID format: **SLUG + random number**, e.g. **SMILE-RWP-01897563**. The slug is generated from the clinic name at setup; the random suffix guarantees uniqueness across all DentOS clinics. This replaces the current auto-generated clinic ID on desktop.

Step B — Prove who you are (two paths)

- **Path 1 · Patient QR:** every patient has a personal QR shown on printed reports/invoices, or the receptionist opens the patient's record on desktop and shows it on screen. Scanning it links instantly — possession of the QR is the proof.
- **Path 2 · CNIC + full name:** patient types both; the backend matches them against that clinic's patient records. Exact match → linked. **CNIC is a mandatory field** on every patient record going forward.

Step C — Session & re-login

- Patient stays signed in on the device; the session **auto-expires after 30 days** (enforced server-side).
- Re-login = same two paths (patient QR or CNIC + name). No SMS, no password to forget.

Security note: patient QR codes carry a signed token (HMAC over clinic_id + patient_uuid), so a guessed or screenshotted ID cannot be forged. CNIC path requires an exact two-field match within the linked clinic only.

3 · App screens (v1)

#	Screen	What it does
1	Welcome / Link clinic	Two options up-front: Scan clinic QR · Enter clinic ID. Confirms clinic name + logo before continuing.
2	Verify identity	Choice of: Scan my patient QR, or CNIC + full name form. Success links the device to the patient record.
3	Home	Next-appointment card, big Book Appointment button, recall banner when a check-up is due.
4	Book appointment	Procedure (clinic's own catalog) -> branch (if multiple) -> dentist (or 'Any') -> date -> free slots -> confirm. Creates
5	My appointments	Upcoming (with cancel / reschedule request) and full visit history with status chips.
6	Profile	Patient's own details, linked-clinic card (call / directions), sign out.
7	Notifications inbox	Booking approved / rejected, 24h & 2h reminders, recall-due alerts.

Bookings are **requests** — the clinic approves or rejects from desktop. The clinic always owns its calendar; patients never hard-book into the schedule. This is the industry-standard model.

4 · Desktop (DentOS) additions

Area	Change
Patients table	New mandatory CNIC column (schema migration + editor field). Existing patients: staff backfill; until then those pa
Patient detail screen	'Show QR' button — displays the patient's signed QR full-screen for scanning at reception; also printed on invoice
Settings -> Patient App card	Shows the clinic QR + clinic ID (SMILE-RWP-01897563 style), with a Print button for reception display.
Clinic ID	Generation changes from plain auto-number to name-slug + random suffix. Set once at setup; shown everywhere
Appointments screen	New 'Requests' inbox badge: pending booking requests with Approve / Reject. Approve creates a real appointment
Sync engine	Pulls booking_requests + patient_accounts; pushes approval results. Same LWW pattern as existing tables.

5 · Backend (Supabase)

New tables

Table	Purpose	Key columns
patient_accounts	One row per linked device/patient	id, clinic_id, patient_uuid, cnic_hash, device_id, session_token, expires_at (30d)
booking_requests	Patient-initiated bookings awaiting clinic approval	clinic_id, branch_id, patient_account_id, procedure, dentist?, requested_slot
clinic_public	Safe, pre-auth clinic info for the link screen	clinic_id, display_name, logo_url, clinic_code (SMILE-RWP-01897563), branches

Edge Functions

Function	Called when	Does
resolve-clinic	Link screen (QR scan or code entry)	QR/code -> clinic_public info; nothing sensitive.
link-patient	Identity step	Verifies patient QR signature OR CNIC+name match within that clinic; mints a 30-day session token.
available-slots	Booking screen	Computes free slots per branch/dentist/date server-side from appointments + clinic schedule.
request-booking	Confirm booking	Validates slot still free; inserts booking_requests; triggers FCM to clinic owner app / desktop.

Security model

- Patient RLS: a patient account can read/write **only its own rows** (auth.uid() match). All clinic data flows through Edge Functions — the app never queries clinic tables directly.
- Patient auth is fully separate from the clinic staff auth model; a compromised patient session exposes only that one patient's data.
- Patient QR tokens are HMAC-signed server-side; CNICs are stored hashed in patient_accounts.
- 30-day expiry enforced in link-patient / every Edge Function call, not just on-device.

Notifications

- Firebase Cloud Messaging (free): booking approved/rejected, reminder 24h and 2h before the appointment, recall-due nudges.
- Triggered by Supabase webhooks / scheduled Edge Functions.

6 · Build phases

Phase	Scope	Effort
P1 · Desktop foundations	CNIC column + editor + migration · clinic-code revamp (slug+random) · patient QR generation + Show QR + print o	S
P2 · Backend	3 tables + RLS · 4 Edge Functions · signing keys · FCM project	M
P3 · Patient app core	Flutter app: link flow (QR/code), identity (QR/CNIC), Home, Book, My Appointments, Profile	
P4 · Requests loop	Desktop Requests inbox + approve/reject -> real appointment · sync engine additions	M
P5 · Notifications & polish	FCM wiring, reminders scheduler, recall nudges, store listings + deep-link web page	M

S = days · M = 1–2 weeks · L = 3–6 weeks (single engineer, familiar stack)

7 · Explicitly OUT of v1

- Invoices / online payments in the app (v2: JazzCash / EasyPaisa deep-links).
- Treatment-plan and prescription viewing (v2).
- Multiple clinics in one app install (v2 — rare case; re-link switches clinic).
- Patient chat / messaging (kept out of scope with WhatsApp policy).

8 · One-time costs & prerequisites

Item	Cost / note
Google Play developer account	USD 25 · one-time
Apple Developer Program	USD 99 / year (can defer; launch Android-first)
SMS / OTP provider	NOT needed — QR / CNIC identity has zero per-login cost
FCM push notifications	Free
Deep-link web page (dentos.app/c/...)	Tiny static page; hosts the clinic QR deferred deep link
Prerequisite	Current desktop branch-isolation work stable and tested before P3 begins

Sign-off: This document reflects the approved flow — clinic link via QR or typed clinic ID (both offered up-front), identity via patient QR or mandatory CNIC + full name (no reception approval step), 30-day sessions, bookings as clinic-approved requests. Next step on approval: Phase P1 (desktop foundations).